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The effect of different charging concepts on hydrogen fuelled internal combustion engines

Onur Barış *, İlker Güler, Anıl Yaşgül

AVL Research and Engineering TR, 34885 Sancaktepe, Istanbul, Turkey

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ABSTRACT

Hydrogen has been widely acknowledged as a potential fuel for internal combustion engines, owing to its unique chemical properties, environmental friendliness, and the ability to be produced from a variety of sources. The carbon-free nature of hydrogen is of particular significance as it is considered to be a key contributor to achieving stringent emissions regulations. However, one of the challenges associated with hydrogen as a fuel is the requirement for high air flow during combustion, which is necessary to meet the stoichiometry of the combustion process. The failure to achieve this high air flow can result in elevated emissions of Nitrogen Oxides (NOx), a phenomenon that is of concern for engine manufacturers.

Turbocharging increases the power and efficiency of internal combustion engines by forcing more air into the combustion chamber. This can be particularly beneficial for hydrogen-fuelled engines as hydrogen has a lower energy density compared to traditional fossil fuels. This study aims to establish the most efficient boosting strategy for hydrogen-fuelled internal combustion engines with the goal of achieving a combination of fuel economy, high power density, and low emissions of NOx. To achieve this objective, a model of a hydrogen-fuelled internal combustion engine, specifically tailored for heavy-duty vehicle applications and featuring a direct injection system, has been developed. The validity of the model has been evaluated through a comparison of the results obtained from the model with those obtained from dynamometer testing. The study employs a model-based approach to examine a variety of charging concepts.

1. Introduction

From the beginning of the 21st century to the next two decades, there has been a significant increase in world population, which indicated a 26% rise as can be seen in [1]. This growth has also led to a demand growth to energy. In order to supply this emerging energy demand, the trend towards carbon-based fuels has resulted in a 39% increase in total Greenhouse Gas (GHG) emissions [2]. Additionally, it is posited that human actions are responsible for global warming, specifically through the rise in GHGs like carbon dioxide in the atmosphere [3]. Analysis of these increased items by sector reveals that industrial usage, electricity and heat production, and transportation are the primary contributors as shown in [2].

In order to decrease the impacts of global warming and ensure a more sustainable environment, reduction of GHG emissions has become a significant concern for the international community. To achieve this goal and comply with strict emission regulations (EU Standard); the automotive industry has also turned to alternative fuels such as electricity, hydrogen, biofuels, ammonia, ether, and ester instead of carbon-based fossil fuels [4–6]. Most scholars and engine manufacturers have highlighted that hydrogen possesses unique physical and chemical properties that make it a more suitable alternative fuel in comparison to other options [7–10]. Even though the ideal reaction of H_2 and O_2 only produces H_2O without any conventional carbon-based emissions (such as CO, CO_2, HC, and soot), the high combustion temperature and the presence of nitrogen gas in the cylinder can lead to the reaction $N_2 + O_2 \rightarrow NOx$. As a result, Hydrogen Fuelled Internal Combustion Engines (H2ICEs) still have the issue of combustion emissions, but these emissions are primarily limited to NOx, which is comparatively simpler to manage than the emissions from other fuels in internal combustion engines [11]. A comparison of the performance of lean-burn without post-treatment, stoichiometric mixture conditions with both Exhaust Gas Recirculation (EGR) and the post-treatment system, revealed that the combination of lean-burn and a turbocharger is the most efficacious method for achieving enhanced efficiency and reduced NOx emissions [12].

Table 1 provides a comprehensive comparison of hydrogen, as a candidate fuel to replace fossil fuels in the future, in relation to its

* Corresponding author.
E-mail addresses: Onur.Baris@avl.com (O. Barış), Ilker.Guler@avl.com (İ. Güler), Anil.Yasgul@avl.com (A. Yaşgül).

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Table 1
Properties of hydrogen compared to gasoline and diesel.

Property	Unit	Gasoline	Diesel	Hydrogen
Density	kg/m^3	760	833	0.09
Lower heating value	MJ/kg	43.5	42.5	120
Stoichiometric air demand	kg_{air}/kg_{fuel}	14.7	14.5	34.3
Mixture calorific value PFI (mixture aspirated)	MJ/m^3	3.76	–	3.19
Mixture calorific value DI (air aspirated)	MJ/m^3	3.83	3.77	4.52
Minimum ignition energy	mJ	0.24	0.24	0.02
Ignition limits	Vol %	1.0–7.6	0.6–5.5	4-76
	λ	0.4–1.4	0.5–1.3	0.2–10
Auto ignition temperature	K	~623	~523	858
Laminar flame speed	cm/s	40–80	40–80	200
Quenching distance	mm	~2	–	0.64
Adiabatic flame temperature	K	2580	~2300	2480
Carbon content	%	84	86	0

combustion properties when compared to conventional fuels such as diesel and gasoline, which are widely utilized in internal combustion engines [13–15].

The alteration in the equivalence ratio had an impact on both the combustion velocity and temperature [11]. Hydrogen has extremely wide ignition limits compared to other conventional fuels which is very well suited for lean operation [14]. Luo [16] divided the change in NOx with equivalence ratio in PFI hydrogen fuel internal combustion engine to four stages. At equivalence ratios less than 0.6, the combustion temperature is the primary factor affecting NOx generation, and the maximum combustion temperature is below 1600 K, which is not sufficient to provide the necessary activation energy, resulting in NOx concentrations of 200 ppm. As equivalence ratios increase to the range of 0.6–0.88, the combustion temperature increases to 1800 K, and the oxygen concentration becomes higher than 4%, creating suitable conditions for NOx generation, leading to NOx concentrations of 7000 ppm which exceed the typical production of NOx in traditional internal combustion engines. As equivalence ratios reach the range of 0.88–1.0, despite the high combustion temperature, NOx emissions rapidly decline to 1000 ppm due to the decrease in oxygen concentration. When the equivalence ratio is higher than 1, the mass concentration of oxygen is close to 0, and the NOx emissions are minimal.

So, the most important factor triggering the formation of NOx emissions is the high temperature reached in the cylinder [17]. Lean mixture formation helps to reduce the combustion temperature thus near zero NOx emission aim can be achieved on H2ICE [8]. Furthermore, one of the key advantages of lean mixture is the improved overall efficiency of the engine, as it eliminates the flow losses (pumping losses) [18].

At stoichiometric engine operation condition, hydrogen has the higher flame speed. This results in a more isochoric, thus thermodynamically more favourable combustion. On the other hand flame speed reduces on the lean mixture condition as explained in [19].

With the highest auto-ignition temperature relative to the common fuels, the resistance of hydrogen to knocking is expected to be high [15]. This feature also proves that hydrogen fuel is more suitable for use in spark ignition engines. However, since the minimum ignition energy value is very low, there is a risk of pre-ignition caused by hot spot if hydrogen fuel is used in spark ignition engines. Also, a non-platinum cold-rated spark plug should be used to avoid risks such as pre-ignition and backfiring in SI engine [20].

Hydrogen, which is distinguished by its combustion characteristics and carbon-free structure, is considered the most viable candidate for the fuel of the future due to its various sources of production and its environmentally friendly production methods [21].

However, on H2ICE, it is vital to determine the correct boosting strategy to keep emissions within regulation limits. According to EURO VI standard, NOX limit is given as 0.4 gr/kWh [22].

In this paper, conventional turbocharging concepts in the automotive sector are considered for Hydrogen fuelled engine. The concepts that are investigated are as follows:

1. Fixed Geometry Turbocharger (FGT)
2. Variable Geometry Turbocharger (VGT)
3. 2-stage Turbocharging (2-stage)

The FGT concept represents the most straightforward and practical solution to implement on an engine. Regardless of whether engine runs on hydrogen or conventional fossil fuels, single stage FGT which uses a wastegate on turbine to control boost pressure by controlling exhaust energy flow passing through the turbine wheel is the cheapest and the simplest turbocharging option compared to single stage VGT and 2-stage turbocharging alternatives [23].

The VGT concept offers a greater degree of flexibility across the entire engine speed range. At the turbine inlet, VGT utilizes racks whose angle can be adjusted with an electronic actuator to control boost pressure. Closing the racks basically creates a smaller turbine which has higher efficiency at low mass flow rates. Moreover, a smaller turbine results in higher turbine inlet pressures at low mass rates, enabling higher energy extraction from exhaust gas by increasing expansion ratio through the turbine. These two advantages of having a smaller turbine are useful for low-end speed region and transient response of the engine [24]. Consequently, VGT provides better performance for wider operation ranges at the expense of complexity and cost due to angle adjustable racks compared to FGT with wastegate option [25].

2-stage turbocharging is the most complex and costly one out of these options. This concept is used when high air flow rates and high boost pressures are required, like hydrogen engines with high air requirements. It has a large turbocharger (LP stage) and a small turbocharger (HP stage) to cover entire engine operating range. As it is more efficient at high mass flow rate operating points, most of the boost pressure demand is satisfied by LP stage at high-end speed region. HP stage with a small turbocharger satisfies most of the boost pressure demand at low-end speed region and is also beneficial for transient response as explained above. HP stage can also be matched as a VGT for better controllability at transient maneuvers. As a result, 2-stage turbocharging can cover a wider operation range with a better transient response compared to single stage VGT [24,26]. However, existence of LP turbine can force HP turbine to operate on low pressure ratio and thereby on low efficiency. Therefore, higher pressure at HP turbine inlet might be required to have the same boost pressure with single stage VGT, which increases pumping losses and consequently fuel consumption compared to single stage VGT option. In addition to that, as it compresses the hot air compressed by LP compressor, HP compressor outlet temperature is likely to exceed hardware limit. To overcome this problem, usage of a titanium compressor or an intercooler between LP and HP compressor is mandatory [26].

In addition to these, more advanced concepts such as e-turbo, e-boost or more advanced electrically assisted turbocharging systems can be modelled. However, these types of concepts are not widely available in the market. In addition, these all systems require higher voltage (48 V or higher), advanced electrical system and/or hybridization. For the sake of system simplicity, these concepts are left out of the scope of this paper, but these can be investigated as a next step if need arises.

Detailed comparison and conclusions of this study will be given in the following parts of the paper but first the details of thermodynamic model will be explained.

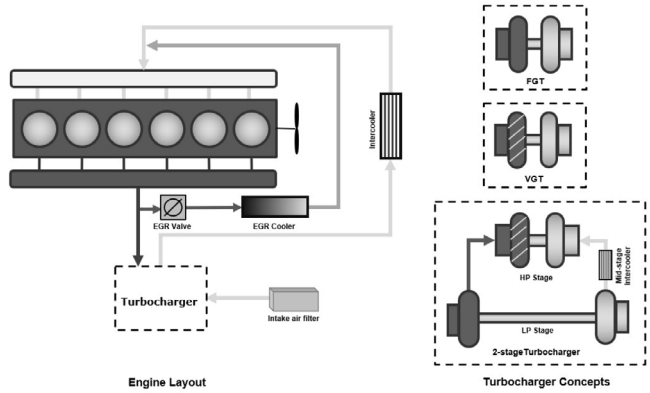

Fig. 1. Engine layout.

Table 2
Engine properties.

Properties	Value
Engine displacement range	12–15L Class
Compression ratio range	9–12
No of cylinders	6
Configuration	Inline
Firing order	1-5-3-6-2-4
EGR system	High Pressure
AirPath	Turbocharger with intercooler

2. Thermodynamic model

In this section, the important model details such as concepts that are selected will be explained. The engine considered here is a heavy-duty inline engine, which is in a displacement widely used in the industry. Fig. 1 shows the schematic view of the engine.

Additionally, the engine properties are indicated in Table 2

Thermodynamical model consists of one-dimensional (1D) model of the air path of the engine. Turbochargers are modelled using the measured performance maps of compressor and turbine parts. Base 1D model was validated with performance data from dynamometer and model deviation was less than ±2% for all test points.

For heat transfer, a flow model is used which considers the squish, swirl, and turbulence flow characteristics inside the cylinder. This model calculates the convective heat transfer within the cylinder according to the flow conditions. The details of this heat transfer model can be found in [27]. A simplistic Finite Element Analysis (FEA) model was used to calculate wall temperature inside the cylinder, where cooling water and oil temperatures are given as boundary conditions.

For the investigation in this paper, low pressure direct injection (LPDI) concept hydrogen engine with spark ignition (SI) is simulated. Therefore, its compression ratio is lower compared to diesel, similar to other heavy duty SI engines such as natural gas (NG) engines. There are also other injection concepts available such as multi-port injection (MPI) which is also spark ignited and high-pressure direct injection (HPDI) with diesel pilot or HPDI hydrogen only (ignited with glow plug or other support system). The reason why LPDI concept is selected because of its best compromise between parameters such as cost, risk, availability (of the injection system) and efficiency [28]. MPI concept is the cheapest among these concepts and easiest to adapt [29]. The injection system is relatively low pressure and less complex. One big disadvantage of MPI is backfire as can be seen in [28]. It is also stated in the same publication that the engine's efficiency is lower in comparison to other concepts. As a result of this lower efficiency, the power density of the engine is also lower in comparison to other concepts. The HPDI with diesel pilot concept is the most similar to the diesel engine concept. The hydrogen is ignited with the means

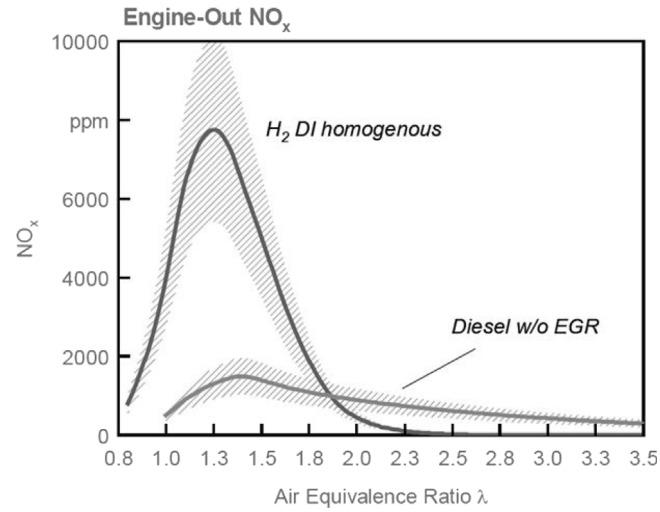

Fig. 2. Typical NOx behaviour of a hydrogen DI engine vs. diesel.

of a diesel pilot injection. Like diesel; the concept benefits from high compression ratio and therefore high thermal efficiency. However, since there is diesel pilot injection, there is also some amount of CO_2 emitted from the engine. Other downside of the engine is the more complex and high pressure fuelling system required for both diesel and hydrogen. Another disadvantage of this system is the high cost related to its high-pressure injection system [28]. Hydrogen only HPDI concept is generally considered to be troublesome and requires a lot of further work as stated in [15,30]. Thus, it is excluded from further consideration. In light of all the information available, it could be deduced that the LPDI presents the best trade-off between all the factors taken into account.

Hydrogen combustion has different emission characteristics compared to diesel, as indicated in [31]. Hydrogen tends to have almost 0 engine-out NOx emission as excess air ratio (λ) goes above 2.4. Fig. 2 [31,32] clearly shows the behaviour of NOx emission as λ changes and it is also stated in the paper explicitly. This will provide the benefit of eliminating the need for a NOx conversion catalyst (such as a selective reduction catalyst-SCR) entirely or using a smaller and simpler one, which implies that the aftertreatment system (ATS) can be less complex and more cost-effective. Therefore, it is chosen to investigate full load performance at high lambda.

In comparison to the engines currently available on the market, this value is exceptionally high and requires a significant amount of air. Therefore, instead of pumping all air into the cylinder, a Cylinder Mass Ratio (CMR) is defined and targeted to 2.4. This is done because EGR is used in the model in all the turbocharging concepts mentioned above. In comparison to λ, CMR is composed of both fresh air and EGR flow, whereas λ only includes fresh air. So, the CMR includes both air and EGR divided by fuel mass and divided by stoichiometric air fuel ratio (AFR) of hydrogen as shown below.

$$CMR = \frac{airflow + EGRflow}{fuelflow * 34.33} \tag{1}$$

As stated in Ref. [33], high heat capacity of EGR will compensate even better than air to lower in cylinder temperatures and therefore also NOx emissions. It is clearly known that NOx is directly related with high temperatures in cylinder as shown in [17]. In addition to these facts, tests are also performed to prove that CMR has similar effect on NOx emission as lambda and this behaviour is shown in Fig. 3.

Since the selected target mass ratio (CMR) is significantly higher than typical diesel and gasoline engines, the question is whether there is a sufficient turbocharger able to supply the air flow demand. It is also the reason why different concepts are investigated in this paper

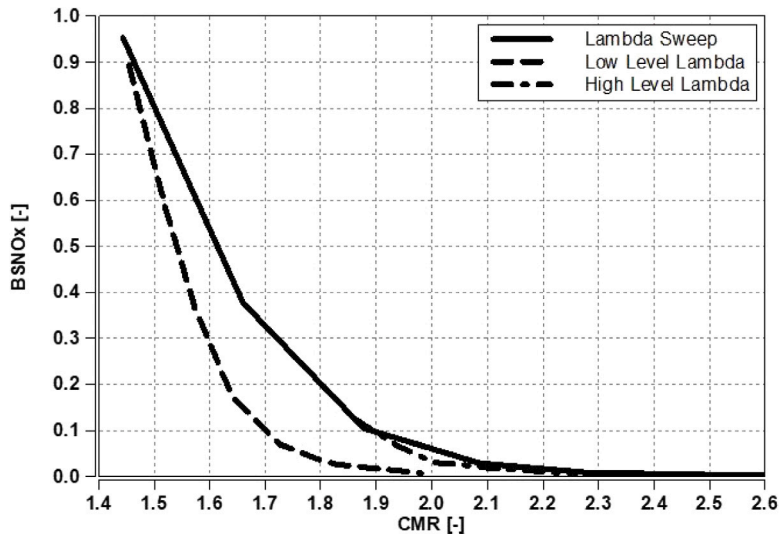

Fig. 3. NOx behaviour comparison including both Lambda and CMR.

in terms of their performance with such high mass ratio, as shown in Fig. 3.

As the choice of the combustion model, SI Wiebe model that obtained from the analyses of test results for each the engine speed and torque is used. Wiebe model is a non-dimensional function to impose burn rate or heat release rate. Wiebe function can be expressed as [34]:

$$x_b = 1 - e^{[-a(\frac{\theta-\theta_0}{\Delta\theta})^{(m+1)}]} \tag{2}$$

In the absence of correlated predictive combustion model, Wiebe function is used for combustion modelling [35]. Since the combustion process occurs within a homogeneous mixture with spark ignition, it is well correlated using SI Wiebe function. In addition, limitations seen in tests are considered when determining the combustion timing to avoid abnormal combustion events due to hydrogens easy ignitability.

As mentioned earlier; EGR system is utilized. A high pressure EGR system between exhaust and intake boost system is used, which is one of the most common concepts. Also, the EGR is cooled because hot EGR will drop the mass flow into the cylinder significantly at high loads. EGR is also beneficial for abnormal combustion events as investigated in [33]. In addition, some of the mass flow required for low NOx is supplied using EGR system, which in turn lowers airflow requirement from the turbocharger. So, the range of airflow required from turbocharger will be lower and a smaller turbocharger can be selected, which will be also beneficial for transient response.

3. Comparison of results

In this section, the comparison of steady state results between different turbocharger concepts will be shown. The main focus of this comparison is to investigate the performance of each turbocharging concept, when targeting low engine-out NOx emissions. Due to this target, the lambda must be lean, around 2.4, which results in a relatively higher demand for air to reach the torque target. In this article, each turbocharger concept is investigated in terms of their response to this challenging air demand. Model results other than brake thermal efficiency (BTE) (Fig. 7), cylinder mass ratio (Fig. 4) and overall TC efficiency (Fig. 8) are plotted in normalized form.

As stated in the thermodynamic model section, this study has been performed by trying to reach CMR target 2.4 and this target has been achieved with all turbocharger concepts as can be seen below. In addition to that, all related results can be seen comparatively between concepts.

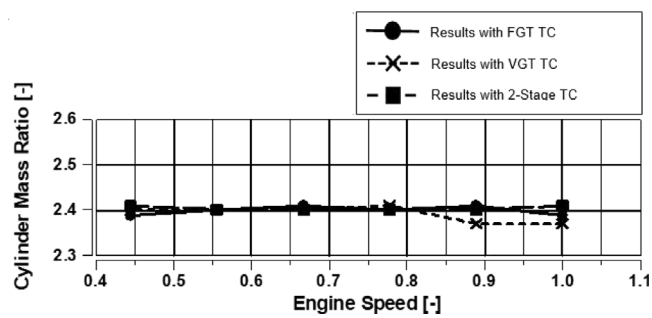

Fig. 4. CMR.

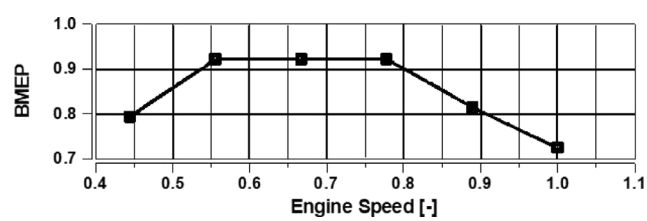

Fig. 5. BMEP.

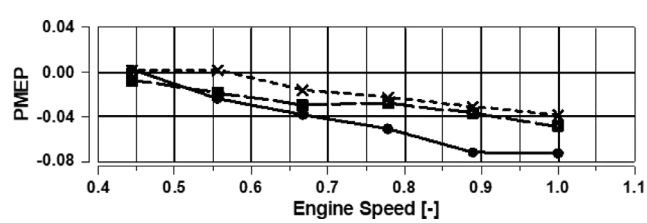

Fig. 6. PMEP.

About Fig. 5, all turbocharger analysis results show same torque output, therefore the graphs are on top of each other, and these cannot be distinguished from each other. About Fig. 10, there is a dashed line added at maximum compressor outlet temperature equals to 1, which indicates the hardware limit. That is a critical value, since the analysis results show that the engine operates close to the limit (see Figs. 4, 6–9 and 11–16).

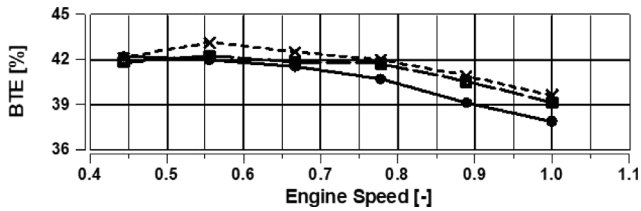

Fig. 7. BTE.

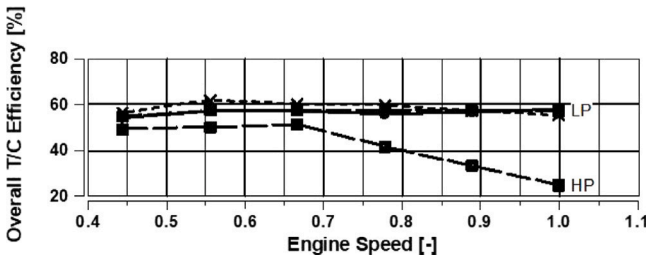

Fig. 8. Overall turbocharger efficiency.

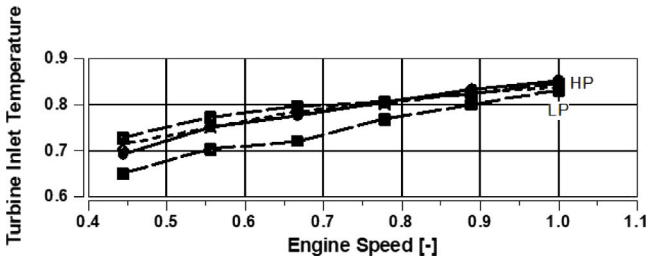

Fig. 9. Turbine inlet temperature.

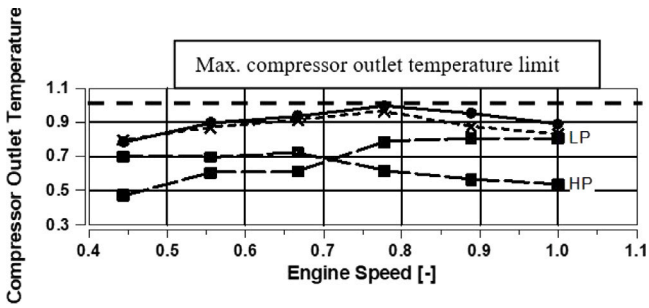

Fig. 10. Compressor outlet temperature.

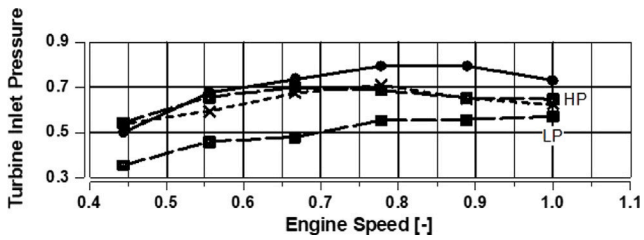

Fig. 11. Turbine inlet pressure.

From BTE point of view, VGT turbocharger has the best efficiency in comparison to FGT and 2-stage configuration and this can be seen in Fig. 7. This can be explained with the PMEP results especially at the low-end-speed region shown in Fig. 6 . FGT has the worst PMEP, and

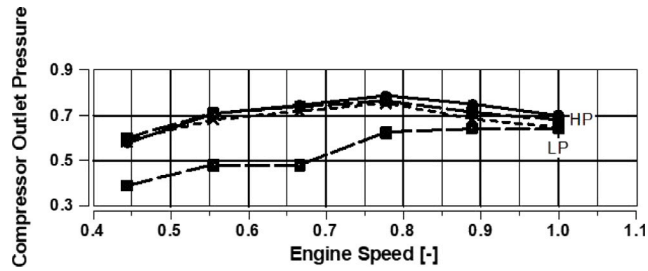

Fig. 12. Compressor outlet pressure.

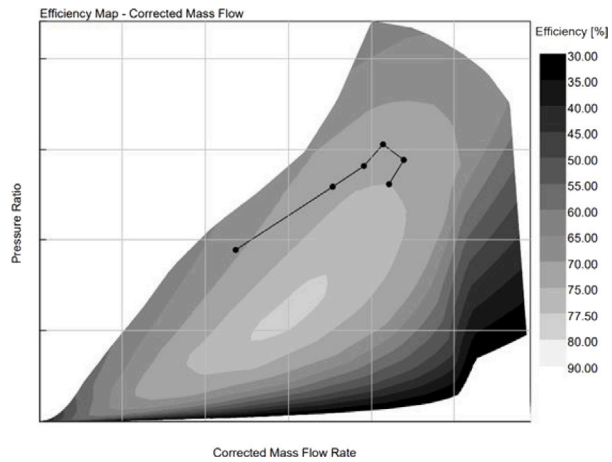

Fig. 13. Compressor map of FGT.

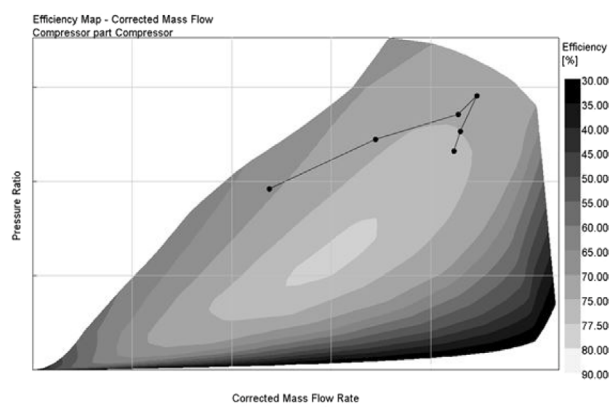

Fig. 14. Compressor map of VGT.

2-stage TC configuration has a PMEP worse than VGT TC configuration at the low-end-speed region. This is because of the size of high-pressure turbocharger of 2-stage charging system is smaller than the size of VGT system. A smaller turbine results in a higher-pressure ratio and higher turbine inlet pressure, which increases pumping work during exhaust stroke.

In addition to that, turbocharger limits are also important parameters to look deeply whether the performance of turbocharger is able to achieve targets at sea level and high-altitude level, too. As can be seen, FGT is already at the limit even at the sea level (compressor outlet temperature please see Fig. 10), because CMR 2.4 target is a hard limit to provide boosted-air when the system has only the waste-gate to satisfy the torque target. Other critical parameters such as turbine inlet temperature and pressure are safely away from their limits, which can be seen in Fig. 9 and Fig. 11, respectively. Similarly compressor outlet pressure shown in Fig. 12 is also much lower than its limit. Their limits

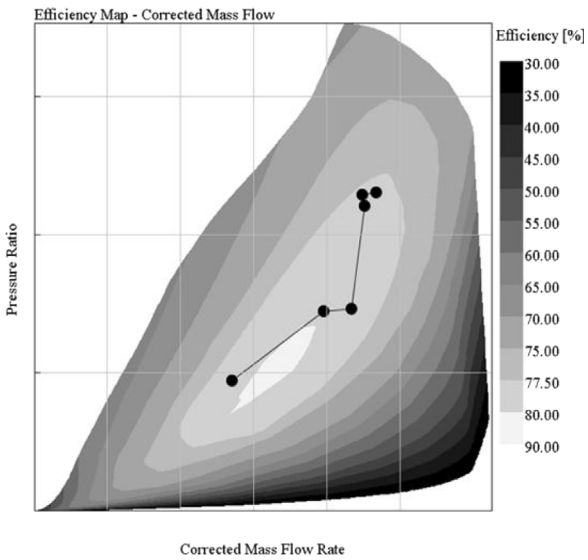

Fig. 15. LP compressor map of 2-stage TC.

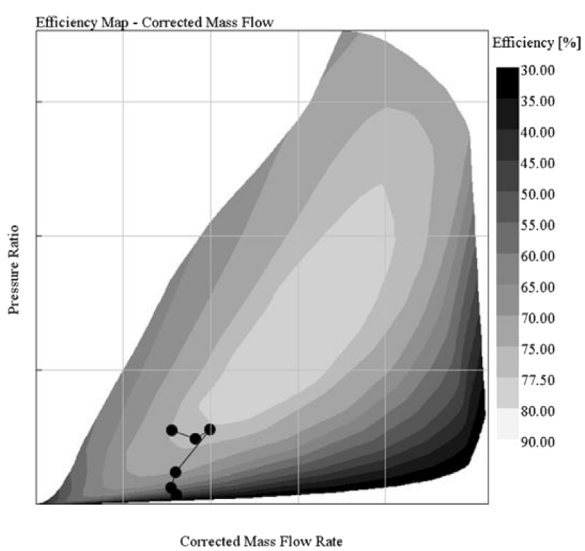

Fig. 16. HP compressor map of 2-stage TC.

are higher than the maximum of y-axis therefore it is not plotted in the figures.

VGT turbocharger system is better in comparison to the system with FGT charging, but this system is also close to the compressor outlet temperature limit. This could make the VGT's performance at high altitude questionable. Compared to FGT; VGT has more margin at higher engine speeds in compressor outlet temperature.

The system with 2-stage turbocharger can be the best option in terms of the high-altitude performance because it has very good margin to stay below the compressor outlet temperature limit in comparison to the systems with FGT and VGT. This means, system with 2-stage charging system can reach higher altitudes with same CMR levels without derating. This is the biggest advantage of 2-stage according to the analysis results.

Turbocharger speed is another important parameter and limit to understand whether turbocharger performance is enough or not for sea level and high altitude. Turbocharger maps are obtained from turbocharger performance test rig. In these tests, turbocharger is tested under certain conditions according to the reference condition and as a result a performance map is obtained. It is a standard turbocharger performance map consisting of the following parameters: pressure ratio, mass flow, efficiency, and turbocharger speed. As can be seen from compressor maps (Fig. 13, Fig. 14, Fig. 15, Fig. 16), all charging concepts have margin to reach the turbocharger speed limit at sea level. VGT has the less margin in terms of turbocharger speed amongst all charging systems, and this means the system with VGT has the least high-altitude potential in terms of turbocharger speed limit. However, in order to have an idea about high altitude potential, turbocharger speed limit and compressor outlet temperature should be considered together.

Table 3
Applications and possible use cases of different charging systems.

	FGT	VGT	2-stage
Genset	X		
Long haul		X	X
Construction			X
Marine	X	X	
Off-road		X	X

Regarding turbocharger efficiencies, it can be said that VGT has the highest efficiencies at most of the operation points. The system with FGT is also close to these efficiency levels and low-pressure turbocharger of 2-stage is also at same levels, where the high-pressure turbocharger of 2-stage looks worse than the others. It is because the low-pressure turbocharger is the more dominant one in terms of having higher compression ratios and that is why the results are closer to the center of low-pressure-charging compressor map. High-pressure stage has lower efficiency because of the lower pressure ratio.

As for transient performance, VGT is expected to respond faster than FGT turbocharger. However, having a smaller turbocharger on high pressure side, 2-stage turbocharging concept is expected to have the best transient response.

4. Conclusion

In this paper, different turbocharging systems are analyzed for hydrogen engines with a very low engine-out NOx emission target. Here instead of lambda, CMR is taken as a target to enable low NOx emission with relaxed air charging requirement.

Following conclusion can be drawn about critical performance criteria:

– Full load target: All turbocharging concepts can achieve the target full load at sea level thanks to the CMR and EGR strategy.

– NOx Emission: Low NOx emissions at steady state conditions are secured by targeting a CMR of 2.4 thanks to the characteristics of hydrogen combustion.

– Efficiency: VGT concept has the best brake thermal efficiency and FGT has the lowest. 2-stage has efficiency close to VGT at higher engine speeds. This is mostly related to pumping losses (PMEP). Low pressure side of the 2-stage turbocharger is more efficient.

– Altitude margin: 2-stage has well enough margin compared to VGT and FGT. FGT has almost no margin and VGT is limited at medium speed in terms of compressor outlet temperature and turbocharger speed. To conclude, 2 stage has the most margin, followed by VGT and FGT, respectively.

From the conclusions drawn here, following applications can be supported with different charging strategies as shown in Table 3.

As a next step, analyses for high altitude and transient load step can be performed for all of the concepts to get a clear picture with solid numbers.

CRediT authorship contribution statement

Onur Barış: Formal analysis, Conceptualization, Investigation, Writing – original draft. **İlker Güler:** Data curation, Draft creation, Supervising, Writing – review & editing. **Anıl Yaşgül:** Formal analysis, Methodology, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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